SIMO Silicon Motion Technology Corporation — 个股深度研究 (专业中文版) v20261016-0756

> 数据截至 2026-10-16 07:56 SGT｜价格 $218.39｜市值 7.4 bn USD｜财报货币 USD｜非投资建议。事实=CONFIRMED/DATA；推断=INFERENCE；假设=ASSUMPTION/HYPOTHESIS。

1. 投资论点 Investment Thesis (HYPOTHESIS)

慧荣是 NAND 周期与 AI 企业级 SSD 的’卖铲人’；收入弹性大，NAND 价格上涨期受益于出货与规格升级。注意最新数据期间 FCF 为负。

- **多头驱动**: Enterprise SSD (MonTitan) wins; PCIe Gen5/6 mix｜企业级 SSD 获单；PCIe 5/6 升级
- **空头驱动**: NAND makers in-source controllers; cycle reverses｜原厂自研主控；周期反转
- **证伪条件 Falsifiers**: Enterprise SSD revenue fails to scale; GM <45%｜企业级 SSD 未放量；毛利率<45%

2. 产品、客户与收入模式

- 主题 Theme: NAND/SSD controllers 存储控制器
- 业务分部 Segments: SSD controllers SSD 主控；eMMC/UFS controllers 嵌入式主控；Other 其他
- 公司描述 (DATA, Yahoo/公司申报): Silicon Motion Technology Corporation, together with its subsidiaries, designs, develops, and markets NAND flash controllers for solid-state storage devices and related devices in China, Japan, Singapore, Taiwan, Korea, the United States, and internationally. It offers controllers for computing-grade solid state drives (SSDs), which are used in PCs and other client devices; enterprise-grade SSDs used in enterprise and hyperscale data centers; eMMC and UFS mobile embedded storage for use in smartphones and IoT devices; flash memory cards and flash drives for use in expandable storage; and specialized SSDs that are used in industrial, commercial, and automotive applications. The company markets its controllers under the SMI brand; and single-chip SSDs under the FerriSSD, Ferri-eMMC, and Ferri-UFS brands. It markets and sells its products through direct sales personnel and independent elect
- 主要客户: Samsung, SK hynix, Micron, Kioxia/SanDisk, YMTC, module makers｜各 NAND 原厂与模组厂

3. 行业、竞争、监管与供应链 (ANALYST FRAMEWORK)

- 供应商/上游: TSMC (controller wafers), OSATs｜台积电、封测厂
- 竞争者 (含业外): Phison, Marvell, Samsung/WD in-house controllers｜群联、Marvell、原厂自研
- 瓶颈/监管: NAND bit growth, enterprise SSD controller design wins｜NAND 位元增长、企业级 SSD 主控订单
- 关键 KPI: Revenue by segment, GM%, enterprise SSD wins｜分业务收入、毛利率、企业级订单
- 零部件现货/合约价: 本版未接入付费价格源(TrendForce/DRAMeXchange 等)——标注 **UNAVAILABLE**，在 KPI 监控中以定性方式跟踪。

4. 历史财务与现金流 (CONFIRMED, 季度, USD)

季度	收入	毛利	营业利润	净利润	FCF	营业利润率
2026-06-30	0.5 bn USD	0.2 bn USD	0.1 bn USD	0.1 bn USD	-71.5 mn USD	22.4%
2026-03-31	0.3 bn USD	0.2 bn USD	52.2 mn USD	66.8 mn USD	-49.4 mn USD	15.3%
2025-12-31	0.3 bn USD	0.1 bn USD	31.7 mn USD	47.7 mn USD	-6.3 mn USD	11.4%
2025-09-30	0.2 bn USD	0.1 bn USD	29.2 mn USD	39.1 mn USD	6.8 mn USD	12.1%
2025-06-30	0.2 bn USD	94.7 mn USD	22.3 mn USD	16.3 mn USD	-32.8 mn USD	11.2%
2025-03-31	N/A	N/A	N/A	N/A	N/A	N/A

资产负债: 现金 74.4 mn USD；总负债 72.5 mn USD；净现金 1.9 mn USD。TTM FCF -0.1 bn USD（FCF 率 -9.2%）。

5. 最新财报: 实际 vs 一致预期 vs 指引

季度	EPS 实际	EPS 一致预期	超预期%
2025-09-30	1.0	0.80667	0.23969999
2025-12-31	1.26	1.29834	-0.0295
2026-03-31	1.58	1.30305	0.2125
2026-06-30	2.43	2.10662	0.1535

- 管理层指引 Guidance: 自动数据源不含指引——**UNAVAILABLE（待从公司 IR 新闻稿录入）**。
- 下次财报日 (Yahoo, TENTATIVE 直至公司确认): **2026-10-29**；本季 EPS 一致预期 3.33339（9 位分析师）；收入一致预期 0.5 bn USD。

6. 未来 2/4/6/8 季度驱动与价格合理性

期限	情景	期限价格现值	相对现价
2Q	Bear	$-13.99	-106.4%
2Q	Base	$-13.64	-106.2%
2Q	Bull	$-4.30	-102.0%
4Q	Bear	$1.78	-99.2%
4Q	Base	$24.47	-88.8%
4Q	Bull	$64.20	-70.6%
6Q	Bear	$17.64	-91.9%
6Q	Base	$67.99	-68.9%
6Q	Bull	$147.64	-32.4%
8Q	Bear	$33.58	-84.6%
8Q	Base	$117.51	-46.2%
8Q	Bull	$248.48	13.8%

现价隐含的 EV/NTM-FCF (Base 路径):	期限
2Q	n/m (neg FCF)
4Q	197.5x
6Q	66.7x
8Q	36.3x

解读: 如果 8 季度后 Base 情景所需倍数仍远高于 25x，说明价格依赖更远期的增长。

7. 三情景 Bear / Base / Bull (HYPOTHESIS)

情景	Y1 增速	长期 FCF 率	WACC	永续 g	概率	每股价值	相对现价
Bear	14.8%	14.4%	14.2%	2.5%	25.0%	$59.08	-72.9%
Base	29.6%	18.0%	13.2%	3.0%	50.0%	$162.44	-25.6%
Bull	38.5%	20.7%	12.7%	3.5%	25.0%	$296.57	35.8%

概率加权价值 **$170.13**（相对现价 -22.1%）。

8. 估值: DCF / Reverse DCF / ROIC-WACC / 同业 / 敏感度

- DCF Base: EV 5.5 bn USD；终值占比 55.5%。
- **Reverse DCF 反向 DCF**: 在 Base 利润率与 WACC 下，现价隐含 Y1 收入增速 ≈ **37.4%**；一致预期 FY+1 增速 29.6%。
- WACC (CAPM) = rf 5.2% + β 1.60 × ERP 5.0% = 13.2%。
- SOTP segments 分部: SSD controllers SSD 主控; eMMC/UFS controllers 嵌入式主控; Other 其他. **Status: framework only** — segment revenues are INPUT REQUIRED from the latest 10-K/10-Q segment note (not auto-fed, left blank to avoid fabrication)｜分部收入需按最新财报手动录入，未自动填充以避免编造。
- 同业比较 Comps (DATA): | Ticker | 市值 | EV/Sales | Fwd P/E | 收入增速 | 毛利率 | FCF 收益率 | |—|—|—|—|—|—|—| | SIMO | 7.4 bn USD | 5.6x | 13.8x | 127.0% | 48.9% | -1.6% | | MU | 1,053.6 bn USD | 7.6x | 4.6x | 379.3% | 80.7% | 2.8% | | MRVL | 216.8 bn USD | 22.5x | 33.1x | 36.5% | 52.2% | 1.1% | | 8299.TWO | 404.7 bn USD | 2.7x | 6.5x | 279.5% | 56.8% | -4.5% |
- 敏感度: 见 Excel `Sensitivity` 表 (WACC × 永续增长率)。

9. 风险、反方论证与证伪

- 反方观点 Counter-thesis: NAND makers in-source controllers; cycle reverses｜原厂自研主控；周期反转
- 证伪指标: Enterprise SSD revenue fails to scale; GM <45%｜企业级 SSD 未放量；毛利率<45%
- 模型风险: 长期利润率与 WACC 为假设；终值占比高时估值对假设极敏感。

10. Source Ledger 来源账本

Item	Value	Source/time
Price 股价	218.4	Yahoo Finance quote, fetched 2026-10-16T07:56 SGT
Market cap 市值	7.405e+09	Yahoo info.marketCap 2026-10-16T07:56
Cash 现金	7.437e+07	Yahoo info.totalCash (latest filing) 2026-10-16T07:56

Item	Value	Source/time
Debt 负债	7.249e+07	Yahoo info.totalDebt (latest filing) 2026-10-16T07:56
Revenue TTM 收入	1.314e+09	Yahoo info.totalRevenue (TTM, USD) 2026-10-16T07:56
FCF TTM 自由现金流	-1.204e+08	Sum of last 4 quarterly FCF (q_cash); quarters ['2026-06-30', '2026-03-31', '2025-12-31', '2025-09-30'] (USD)
Beta	1.682	Yahoo info.beta (5Y monthly)
Risk-free 10Y 无风险利率	0.0519	FRED DGS10 2026-10-14
FX USD per USD	1	n/a (USD)
Consensus rev growth FY0	1.117	Yahoo revenue_estimate 0y
Consensus rev growth FY+1	0.2964	Yahoo revenue_estimate +1y

11. Assumption Register 假设登记表

Assumption	Value	Rationale
ERP 股权风险溢价	0.05	ASSUMPTION: 5.0% house mature-market ERP
Beta clamp	[0.8,1.6] → 1.60	ASSUMPTION: limit extreme betas
Tax rate (ROIC)	0.18	ASSUMPTION: blended cash tax 18%
Year-1 growth (Base)	0.2964	Consensus FY+1 revenue growth (Yahoo revenue_estimate '+1y'); capped at 100%
Growth fade	linear Y1→terminal by Y10	ASSUMPTION
Starting FCF margin	-0.0917	DATA: FCF TTM / Revenue TTM, floored -50%

Assumption	Value	Rationale
Long-run FCF margin (Base)	0.18	ASSUMPTION (analyst framework, reached by Y5; see profile)
Bear/Bull Y1 growth multiplier	0.5x / 1.3x	ASSUMPTION
Bear/Bull margin multiplier	0.8x / 1.15x	ASSUMPTION
WACC adj Bear/Bull	+1.0pp / -0.5pp	ASSUMPTION
Terminal g Bear/Base/Bull	2.5% / 3.0% / 3.5%	ASSUMPTION
Scenario probabilities	25% / 50% / 25%	ASSUMPTION
Horizon exit EV/NTM-FCF Bear/Base/Bull	15x / 25x / 35x	ASSUMPTION
Per-share conversion	value × fx ÷ market cap × price	METHOD: avoids ADR/share-class mismatch

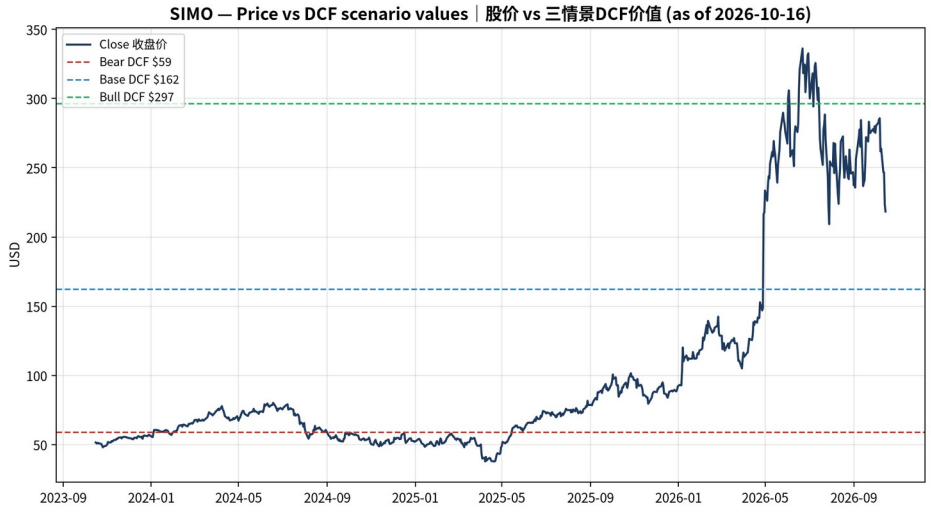
12. Model Provenance 模型溯源

- Model file: `SIMO_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions)
- Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx)
- Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT
- Version: v20261016-0756. Prior versions are never overwritten (see company history page).

13. 版本与事件时间轴

- v20261016-0756: 首版生产系统报告 (27 家同步)。下一事件: 财报 2026-10-29 (TENTATIVE)。

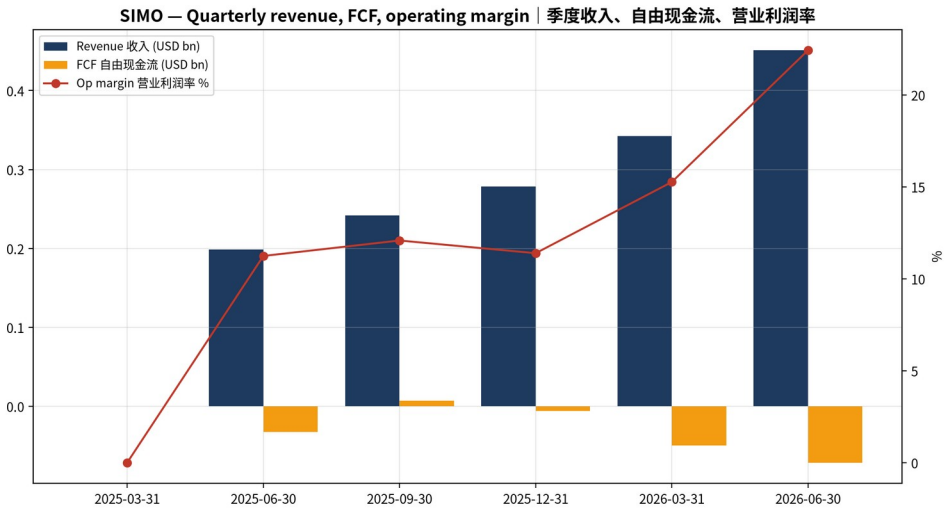
图片报告 Image Report



Claim class 声明类别: DATA (price) + HYPOTHESIS (scenario values) | Source 来源: Yahoo Finance daily closes; model SIMO_model.xlsx
Thinking 思考: Gap between price and Base line = what the market believes beyond our base assumptions → see Reverse DCF. | Not investment advice 非投资建议

1_price_vs_value

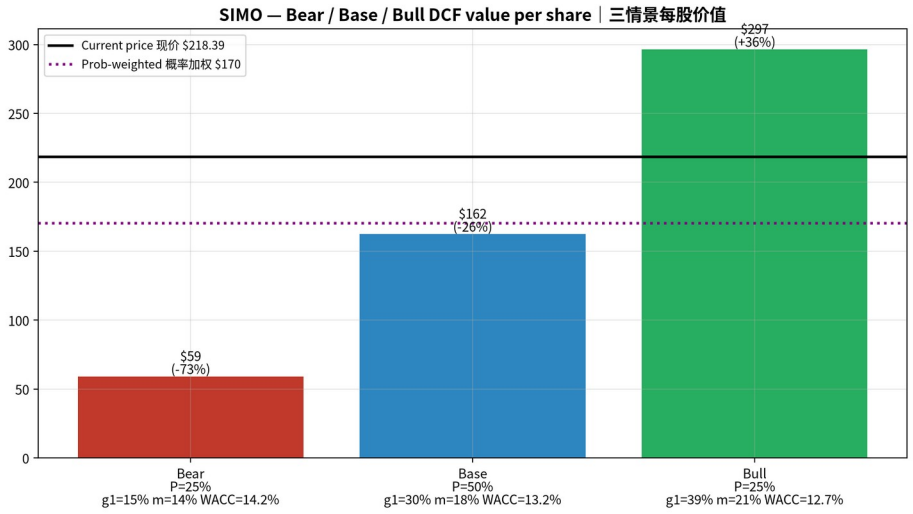
股价vs 三情景价值Price vs scenario values



Claim class 声明类别: CONFIRMED (reported) | Source 来源: Company filings via Yahoo Finance quarterly statements
Thinking 思考: Is FCF growing with revenue? Divergence = capex/working-capital intensity → check cash-flow statement. | Not investment advice 非投资建议

2_financials

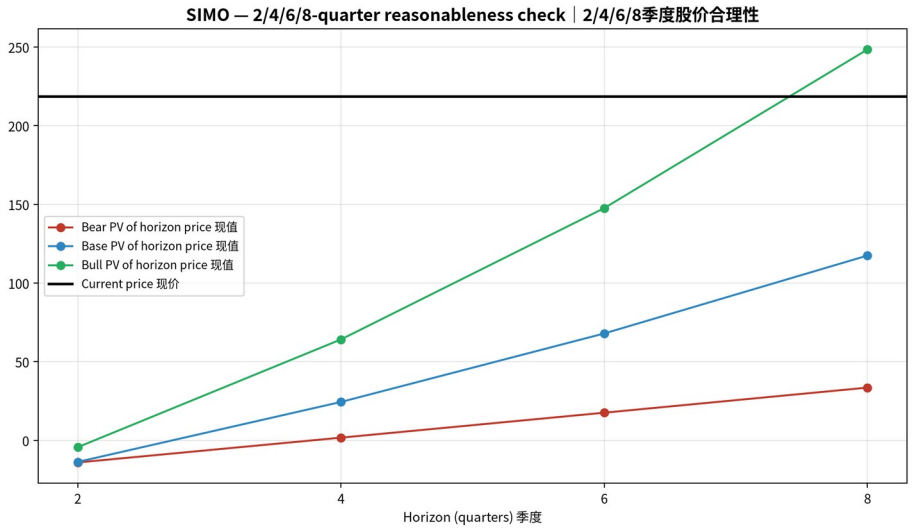
季度财务 Quarterly financials



Claim class 声明类别: HYPOTHESIS (model output) | Source 来源: Model (live formulas) — inputs in Sources/Assumptions sheets
Thinking 思考: Which assumption moves value most? Check Sensitivity sheet; scenario width = uncertainty. | Not investment advice 非投资建议

3_scenarios

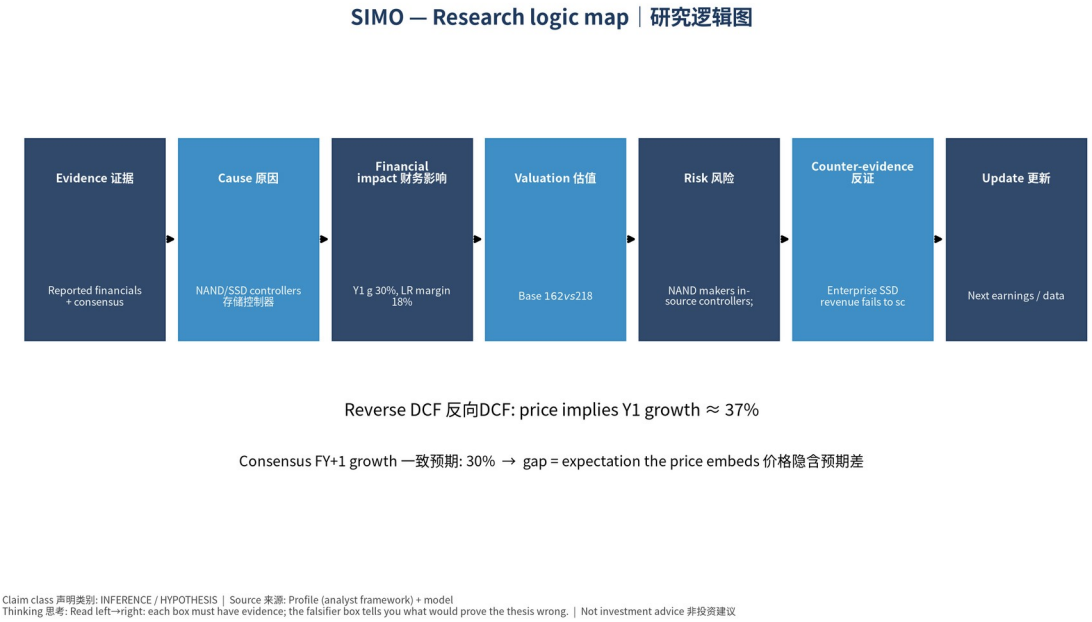
三情景 Scenarios



Claim class 声明类别: HYPOTHESIS | Source 来源: Model Horizons sheet; exit EV/NTM-FCF 15x/25x/35x (ASSUMPTION)
Thinking 思考: If even Bull PV < price at 8Q, price requires assumptions beyond Bull within 2 years. | Not investment advice 非投资建议

4_horizons

2/4/6/8 季度检验 Horizon check



5_logic_map

研究逻辑图Logic map

SIMO Silicon Motion Technology Corporation — Equity Deep Dive (Professional English) v20261016-0756

> Data as of 2026-10-16 07:56 SGT | Price $218.39 | Mkt cap 7.4 bn USD | Reporting currency USD | Not investment advice.

1. Investment thesis (HYPOTHESIS)

Silicon Motion is a ‘picks-and-shovels’ play on the NAND cycle and AI enterprise SSDs; revenue is highly elastic and benefits from NAND upcycles and spec upgrades. Note negative FCF in latest data period.

- **Bull drivers**: Enterprise SSD (MonTitan) wins; PCIe Gen5/6 mix｜企业级 SSD 获单；PCIe 5/6 升级
- **Bear drivers**: NAND makers in-source controllers; cycle reverses｜原厂自研主控；周期反转
- **Falsifiers**: Enterprise SSD revenue fails to scale; GM <45%｜企业级 SSD 未放量；毛利率<45%

2. Products, customers, revenue model

- Theme: NAND/SSD controllers 存储控制器 | Segments: SSD controllers SSD 主控; eMMC/UFS controllers 嵌入式主控; Other 其他
- Business description (DATA): Silicon Motion Technology Corporation, together with its subsidiaries, designs, develops, and markets NAND flash controllers for solid-state storage devices and related devices in China, Japan, Singapore, Taiwan, Korea, the United States, and internationally. It offers controllers for computing-grade solid state drives (SSDs), which are used in PCs and other client devices; enterprise-grade SSDs used in enterprise and hyperscale data centers; eMMC and UFS mobile embedded storage for use in smartphones and IoT devices; flash memory cards and flash drives for use in expandable storage; and specialized SSDs that are used in industrial, commercial, and automotive applications. The company markets its controllers under the SMI brand; and single-chip SSDs under the FerriSSD, Ferri-eMMC, and Ferri-UFS brands. It markets and sells its products through direct sales personnel and independent elect
- Customers: Samsung, SK hynix, Micron, Kioxia/SanDisk, YMTC, module makers｜各 NAND 原厂与模组厂

3. Industry, competition, regulation, supply chain (ANALYST FRAMEWORK)

- Suppliers: TSMC (controller wafers), OSATs｜台积电、封测厂
- Competitors: Phison, Marvell, Samsung/WD in-house controllers｜群联、Marvell、原厂自研
- Bottlenecks/regulation: NAND bit growth, enterprise SSD controller design wins｜NAND 位元增长、企业级 SSD 主控订单
- KPIs: Revenue by segment, GM%, enterprise SSD wins｜分业务收入、毛利率、企业级订单
- Component spot/contract prices: paid feeds not connected — **UNAVAILABLE**, tracked qualitatively.

4. Historical financials & cash flow (CONFIRMED, quarterly, USD)

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
2026-06-30	0.5 bn USD	0.2 bn USD	0.1 bn USD	0.1 bn USD	-71.5 mn USD	22.4%
2026-03-31	0.3 bn USD	0.2 bn USD	52.2 mn USD	66.8 mn USD	-49.4 mn USD	15.3%

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
2025-12-31	0.3 bn USD	0.1 bn USD	31.7 mn USD	47.7 mn USD	-6.3 mn USD	11.4%
2025-09-30	0.2 bn USD	0.1 bn USD	29.2 mn USD	39.1 mn USD	6.8 mn USD	12.1%
2025-06-30	0.2 bn USD	94.7 mn USD	22.3 mn USD	16.3 mn USD	-32.8 mn USD	11.2%
2025-03-31	N/A	N/A	N/A	N/A	N/A	N/A

Cash 74.4 mn USD; debt 72.5 mn USD; TTM FCF -0.1 bn USD (FCF margin -9.2%).

5. Latest earnings: actual vs consensus vs guidance

Quarter	EPS actual	EPS consensus	Surprise
2025-09-30	1.0	0.80667	0.23969999
2025-12-31	1.26	1.29834	-0.0295
2026-03-31	1.58	1.30305	0.2125
2026-06-30	2.43	2.10662	0.1535

- Management guidance: not in automated feed — **UNAVAILABLE (to be entered from IR release)**.
- Next earnings (Yahoo; TENTATIVE until company-confirmed): **2026-10-29**; current-quarter EPS consensus 3.33339 (9 analysts); revenue consensus 0.5 bn USD.

6. 2/4/6/8-quarter drivers & price reasonableness

Horizon	Scenario	PV of horizon price	vs current
2Q	Bear	$-13.99	-106.4%
2Q	Base	$-13.64	-106.2%
2Q	Bull	$-4.30	-102.0%
4Q	Bear	$1.78	-99.2%
4Q	Base	$24.47	-88.8%
4Q	Bull	$64.20	-70.6%
6Q	Bear	$17.64	-91.9%
6Q	Base	$67.99	-68.9%
6Q	Bull	$147.64	-32.4%
8Q	Bear	$33.58	-84.6%

Horizon	Scenario	PV of horizon price	vs current
8Q	Base	$117.51	-46.2%
8Q	Bull	$248.48	13.8%

Current-price implied EV/NTM-FCF on Base path:	Horizon
2Q	n/m (neg FCF)
4Q	197.5x
6Q	66.7x
8Q	36.3x

7. Bear / Base / Bull (HYPOTHESIS)

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Bear	14.8%	14.4%	14.2%	2.5%	25.0%	$59.08	-72.9%
Base	29.6%	18.0%	13.2%	3.0%	50.0%	$162.44	-25.6%
Bull	38.5%	20.7%	12.7%	3.5%	25.0%	$296.57	35.8%

Probability-weighted value **$170.13** (-22.1% vs price).

8. Valuation: DCF / Reverse DCF / ROIC-WACC / comps / sensitivity

- Base DCF EV 5.5 bn USD; terminal share 55.5%.
- **Reverse DCF**: at Base margin/WACC the price implies Y1 revenue growth ≈ **37.4%** vs consensus FY+1 29.6%.
- WACC (CAPM) = rf 5.2% + β 1.60 × ERP 5.0% = 13.2%.
- SOTP segments: SSD controllers SSD 主控; eMMC/UFS controllers 嵌入式主控; Other 其他 — framework only; segment revenue inputs pending from filings.
- Comps (DATA): see table in Chinese section / Excel `Comps`.
- Sensitivity: Excel `Sensitivity` (WACC × terminal g).

9. Risks, counterarguments, falsifiers

NAND makers in-source controllers; cycle reverses｜原厂自研主控；周期反转 | Falsifiers: Enterprise SSD revenue fails to scale; GM <45%｜企业级 SSD 未放量；毛利率<45%

10–12. Source Ledger / Assumption Register / Model Provenance

See tables above (shared, bilingual). - Model file: `SIMO_model_v20261016-0756.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions) - Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx) - Data snapshot: `data/snapshots/20261016-0756` (immutable) | Fetched 2026-10-16 07:56 SGT | Generated 2026-10-16 07:57 SGT - Version: v20261016-0756. Prior versions are never overwritten (see company history page).